

RMO(INDIA) – 2009

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let ABC be a triangle in which $AB = AC$ and let I be its in-centre. Suppose $BC = AB + AI$. Find $\angle BAC$. [16]
 2. Show that there is no integer 'a' such that $a^2 - 3a - 19$ is divisible by 289. [16]
 3. Show that $3^{2008} + 4^{2009}$ can be written as product of two positive integers of which is larger than 2009^{182} . [15]
 4. Find the sum of all 3 digit natural numbers which contain at least one odd digit and atleast one even digit. [15]
 5. A convex polygon Γ is such that the distance between any two vertices of Γ does not exceed 1. [19]
 - (i) Prove that the distance between any two points on boundary of Γ does not exceed 1.
 - (ii) If X and Y are two distinct points inside Γ , prove that there exist a point Z on the boundary of Γ such that $XZ + YZ \leq 1$.
 6. In a book with page numbers from 1 to 100, some pages are torn off. The sum of the numbers on the remaining pages is 4949. How many pages are torn off? [19]

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